

Brain Stroke Classification using One Dimensional Convolutional Neural Network

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Abstract—Advances in technology have now developed into various fields, one of which is the health sector. With the support of this technology, it can help the work of health workers in dealing with their patients. Not only acting as a curative and rehabilitative, but technology can also be used as a preventive medium for dangerous diseases; an example is a stroke. According to the World Health Organization (WHO), stroke is the second leading cause of death and the third leading cause of disability. Stroke can occur due to lifestyle factors (such as tobacco and alcohol use) and medical factors (such as a history of heart disease and hypertension). Data on patients who have factors that trigger stroke can be used to detect and predict the potential for the patient to have a stroke or not. The research was conducted using a Deep Learning algorithm, namely Convolutional Neural Network (CNN), to train the data to produce an architecture that can predict the possibility of brain stroke. The dataset used in this study was obtained from the Brain Stroke Detection public domain dataset, which has 11 attributes (including gender, age, hypertension, heart disease, ever married, work type, residence type, average glucose level, BMI, smoking status, and stroke). The architecture produced an excellent accuracy of 98%, an F1-Score of 98%, and a Loss of 0.1180. Besides, this architecture has a good fitting, which shows that the architecture made is robust enough to predict the potential for the occurrence of brain stroke.

Index Terms—brain stroke, classification, convolutional neural network

I. INTRODUCTION

Stroke is one of the deadliest diseases in the world. In 2016 stroke was ranked 2nd in the disease that causes death under heart disease [1]. Meanwhile, in 2019 there were 12.2 million stroke cases with a global prevalence of 101 million. In addition, in that year, 6.6 million cases of death were caused by stroke [2], [3].

Stroke or Cerebrovascular Accident (CVA) is a medical term to describe the brain and spinal cord injury due to the abnormal blood supply to the brain [4], [5]. Stroke is divided into three types, and there is Ischemic Stroke (IS), Hemorrhagic Stroke (HE), and Transient Ischemic Attack (TIA) [4]. As many as 85% of stroke cases are ischemic stroke types [1]. *Ischemic stroke* is a stroke caused by a blockage of blood flow. Bursts of blood flow cause a hemorrhagic stroke (bleeding in the brain) [4]. While TIA

is a temporary disturbance in blood flow and can also be a warning that a stroke will occur soon [4], [6].

People aged of 65 years and over have a high enough risk for stroke [7]. Five factors cause stroke: high blood pressure, high BMI, high blood glucose levels, high ambient particulate matter pollution, and smoking habits [3]. Diagnosing stroke can be done by several physical and neurological examinations and brain imaging. Brain imaging can usually be a Computed Tomography (CT) scan or Magnetic Resonance Imaging (MRI) [1].

To find out the prediction of stroke, it costs quite a lot to carry out several health tests. In addition, a stroke is usually recognized when a stroke has been diagnosed. However, this can be anticipated by conducting early detection based on tests regarding the factors that cause a stroke. Therefore, in this study, we will discuss the early prediction of stroke based on test data regarding the factors that cause stroke that is easily obtained.

Priya G, et al. [4] studied the Classification of Stroke Disease using Machine Learning Algorithms, which classify strokes by combining text mining and machine learning algorithms. The dataset contains data on 507 stroke patients in the age range of 35-90 years. This study uses an Artificial Neural Network (ANN) model with 22 input layers and a hidden layer. Support Vector Machine (SVM) for Linear SVM, Quadratic SVM, Cubic SVM, Fine Gaussian SVM, Medium Gaussian SVM, and Coarse Gaussian SVM. Decision Tree, Logistic Regression, AdaBoost, and Random Forest. The ANN model obtained the highest accuracy, with an accuracy of 95.3%.

Yifeng Xie et al. [8] studied stroke prediction from electrocardiograms by Deep Neural Network using the ECG dataset. The dataset contains 100 360 Hz ECG samples with 12 standard lead from people with normal conditions and stroke patients. They use the DenseNet model with 12 layers and reach 85.82 % accuracy. However, the accuracy is not high enough. Faisal, A. and Subekti, A. [5] conducted a study on a Deep Neural Network for stroke prediction consisting of 5110 instants with 12 attributes. The model used is Deep Neural Network (DNN), and the accuracy is 96%. However,

the dimensions of the dataset are not significant. Emon, MU et al. [9] researched Performance Analysis of Machine Learning Approaches in Stroke Prediction with various models such as LR, SGD, DTC, AdaBoost, Gaussian, QDA, MLP, KNeighbors, GBC, XGB, and Weighted. The model with the highest accuracy is weighted voting with an accuracy of 97%.

Gaidhani BR et al. [10] discusses Brain Stroke Detection using Convolutional Neural Network and Deep Learning Models using MRI images as the dataset. The dataset contains 229 brain stroke anatomical images and 400 normal brain anatomical images. This study uses a CNN model, which consists of 2 types: LeNet and SegNet and deep learning. The accuracy obtained by the classification model is 96-97%, while the accuracy of the segmentation model is 85-87%. However, the dataset used is small. Liu, T., et al [11] conducted a study on A Hybrid Machine Learning Approach to Cerebral Stroke Prediction Based on Imbalanced Medical Dataset using the DNN model and achieved an accuracy of 71.67%. It can be seen that the accuracy is still relatively low. Monteiro, M., et al [12] conducted a study on Using Machine Learning to Improve the Prediction of Functional Outcome in Ischemic Stroke Patients by adding several features to increase the AUC value. The dataset contains 152 features and 426 patients and has an increased AUC value of 0.936.

Li, X., et al [13] conducted a study on Using Machine Learning Models to Improve a Stroke Risk Level Classification Methods of China National Stroke Screening by comparing several existing models. The dataset used is the result of screening for stroke in 2017 in China, with a total of 747,514 participants. Using the boosting model with decision trees, the highest recall value is 99.94%, and the precision value from the random forest model is 97.33%. Liu, Y., et al [14] conducted a study on The Probability of Ischemic Stroke Prediction with a Multi-Neural-Network Model using a dataset on cerebral vasoregulators in the elderly with stroke. The model used is CNN based on VGG16, with prediction accuracy increasing to 98.53%. Singh, MS, and Choudhary, P. [7] conducted a study on Stroke Prediction using Artificial Intelligence. The Cardiovascular Health Study (CHS) dataset studies heart disease and stroke in adults aged 65 years and over. This dataset contains 5,888 samples consisting of males and females with 600 attributes. A decision tree is used for feature selection and Neural Network Classification to create a classification model so that an accuracy value of 97.7% is obtained.

This study uses a stroke prediction dataset which consists of 9 features with a total of 4981 data. However, the dataset is imbalanced. To overcome this problem, the RandomOverSampler and StandardScaler functions are used to standardize the data. The model used is Convolutional Neural Network with four convolutional layers and two hidden layers. Using Adam Optimizer, the learning rate is 0.001, decay is 0.000001, and the total epochs are 120. The model accuracy is 98%. Compared with similar previous studies, this study has a better accuracy value using the CNN model.

This research produce architecture that can predict the potential of a person who suffer from stroke, with Conv1D for classification based on several features that has correlation with brain stroke.

Next, Chapter II describes the method used to classify brain

stroke, which consists of data collection and data processing. Chapter III describes the results and discussion. Then in chapter IV contains the conclusion.

II. METHOD

The general method consists of several steps: dataset collection, data preprocessing, model training, and model testing. The method is generally shown in figure 1.

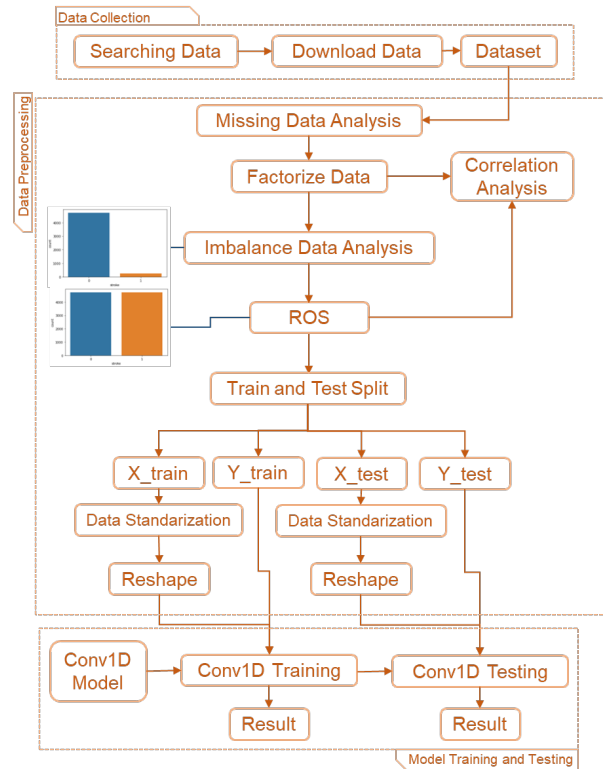


Fig. 1. Research Model Workflow

A. Data Collection

Brain Stroke datasets were downloaded from the website named Kaggle. The following attributes and the information about the attributes are explained below.

- gender : “Male”, “Female”, and, “Other”
- age : age of the patient
- hypertension : 0 if the patient doesn’t have hypertension, 1 if the patient has hypertension.
- heart_disease : 0 if the patient doesn’t have heart disease, 1 if the patient have heart disease.
- ever_married : “No” or “Yes”
- work_type : “children”, “Govt_job”, “Never_worked”, “Private”, or “Self-employed”
- Residence_type : “Rural” or “Urban”
- avg_glucose_level : average glucose level in blood.
- bmi : body mass index
- smoking_status : “formerly smoked”, “never smoked”, “smokes”, atau “Unknown”
- stroke : 1 if patient had a stroke or 0 if not.

There is a note for smoking_status attribute that the “Unknown” value means that data is unavailable.

B. Data Preprocessing

Before the data is inputted into the model, data must be analyzed and cleaned first so the model will be trained and tested well. The first step is we find the information about the data, like unknown data, the amount of value in each data attribute, information about imbalanced data, and the correlation between one data attribute and the others. Data information are visualized in figure 2.

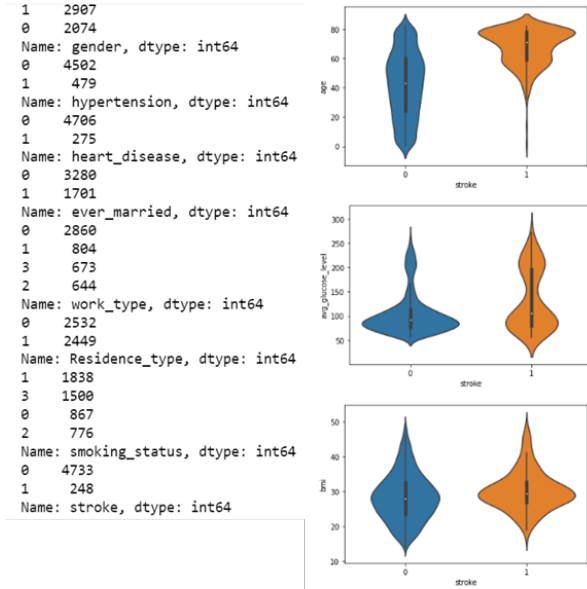


Fig. 2. Each data feature value information

Unknown data are only found in the “stroke” attribute. Of 4981 data, there are 1500 unknown data, or it is 30% of the data unknown. The unknown data is not deleted because it is too much. Then the data was analyzed with some python functions and seaborn visualization. Countplot is used for visualizing the imbalance target attribute that is “stroke” and heatmap is used for visualizing the correlation between each data attribute. Imbalance stroke is shown in figure 3 and data correlation is shown in figure 4.

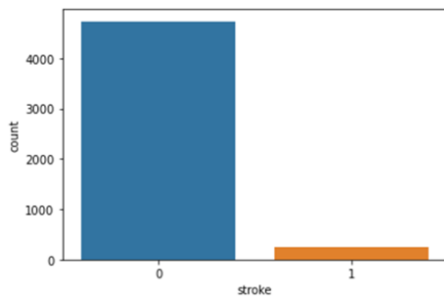


Fig. 3. Diagram of stroke value before using random oversampler

The random oversampling method is used to balance the target attribute. After a random oversampling is used, the “stroke” attribute is balanced, as shown in figure 6. After the dataset is balanced, we visualize each data’s correlation. After the data is repaired, data is divided into data for training and data for testing using train_test_split() function. For testing, we use 20% of the data and 80% of the data allocated for training. Then the data feature for training and testing

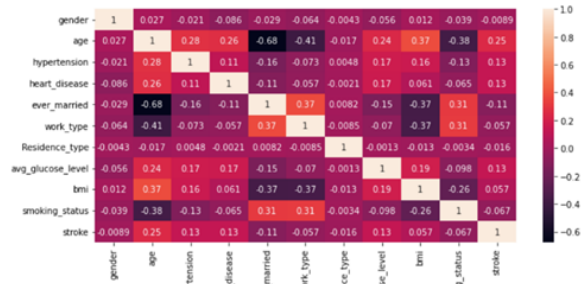


Fig. 4. Heatmap of each other data correlation before using random oversampler

is standardized by using the standard scaler function from sklearn. The feature of train data reshaped from (7572, 10) into (7572, 10, 1), and the feature of test data reshaped from (1894, 10) into (1894, 10, 1). Visualization of balanced stroke values are shown in figure 5 and data correlation is shown in figure 6.

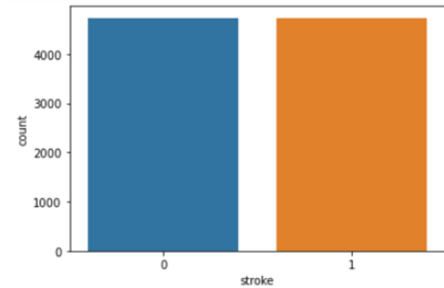


Fig. 5. Diagram of stroke value after using random oversampler

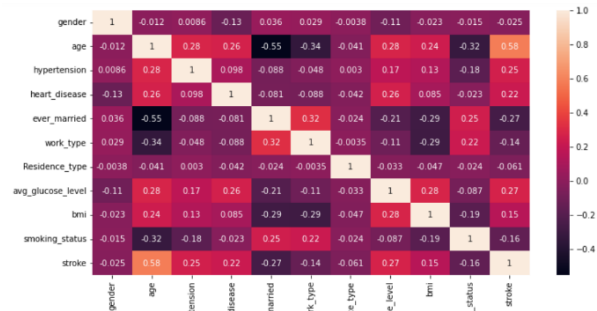


Fig. 6. Heatmap of each other data correlation after using random oversampler

C. Convolutional Neural Network Modelling

CNN from the Keras library is used with several functions that are **Conv1D()** for the convolutional layer, **MaxPooling1D()** for the max pooling layer, **Dropout()** is used to prevent overfitting, **Flatten()** to change the data into 1 dimension shape, and **Dense()**. Our CNN architecture is shown in figure 7.

First, the data was convoluted two times. The first convolution uses four filters with two filter sizes, and the second convolution uses 12 filters with three filter sizes. Then the

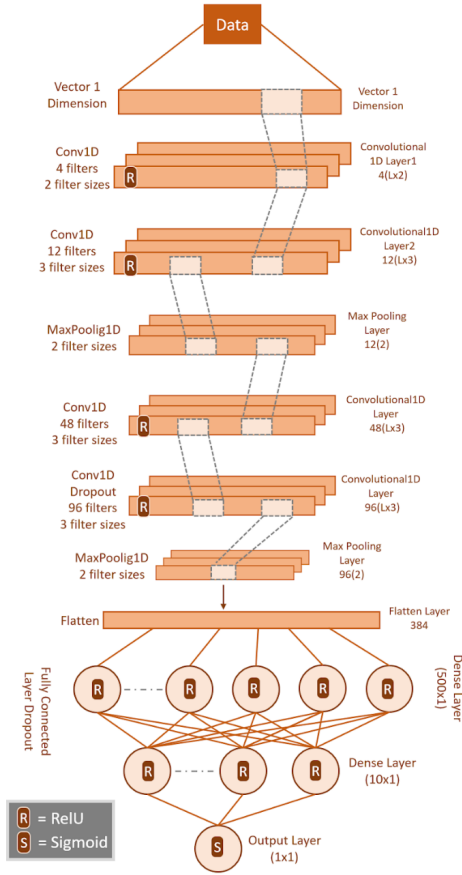


Fig. 7. Architecture of CNN model

data were pooled using maxPooling by taking the largest value with two filter sizes. After that, the data convoluted again two times, where the first convolution uses 48 filters with three filter sizes, and the second convolution uses 96 filters with three filter sizes. Then the data were pooled again by taking the largest value with two filter sizes. After that data flatten, the data proceed into the neural network where the first layer is the dense layer with 500 nodes, the second layer is the dense layer with ten nodes, and the third layer is the output layer with one node. The convolution and dense layer use the ReLU activation function, and the output layer uses the Sigmoid activation function.

III. RESULT AND DISCUSSION

This research will contribute in health sector in early prevention from the possibility of a person suffer from brain stroke. By using several features that have correlation with the cause of stroke, architecture that we made can predict the potentation of a person suffer from a stroke. In addition, this architecture can be developed to be applied to health applications, websites, and others, so that it can be more easier used by wide community.

A. Dataset Summary

Figure 8 represents the correlation between each feature in the dataset that has been factored using a heatmap graph. The closer the correlation between each feature is indicated

by the larger the number and the brighter the color. This graph shows the correlation between essential features (i.e., gender, age, hypertension, Etc.) and target features (stroke). If viewed from the graph, the highest correlation to stroke is age, with a correlation of 0.25, followed by hypertension, heart disease, and average glucose level, with a correlation of 0.13.

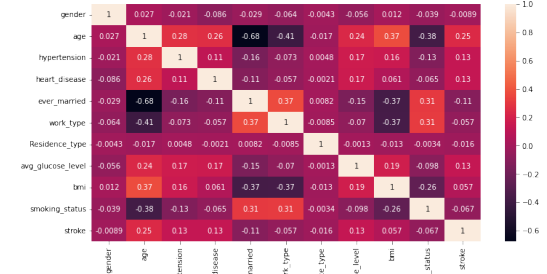


Fig. 8. Correlation between each feature after factorized

B. Architectural Evaluation

Based on the architecture, the resulting accuracy is very satisfactory, which is 98%. It can be seen in the accuracy graph in Figure 9, the results of train and validation data accuracy both increase in line with the epoch. This architecture also produces a reasonably good loss, as seen in the loss chart in Figure 10. It can be said that this architecture is robust.

The architecture created is then evaluated using standard performance metrics, namely by calculating the accuracy, precision, recall, and F1-score values. Accuracy is a measure of how much the architecture is made to identify all positive and negative cases correctly, as shown in equation 1

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \times 100\% \quad (1)$$

Precision is the architectural ability to correctly categorize positive samples from all positive cases, as shown in equation 2.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

A recall is the architectural ability to correctly categorize positive samples from all positive and negative cases, as shown in equation 3.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

F1-Score is a weighted average of precision and recall, as shown in equation 4:

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \times 100\% \quad (4)$$

The architecture created has a very satisfying precision, namely 96% for healthy people and 99% for people who are indicated to have a stroke. In addition, this architecture produces 98% for F1-score.

In addition, the performance evaluation of the model is represented as a confusion matrix, as shown in Figure 11. It

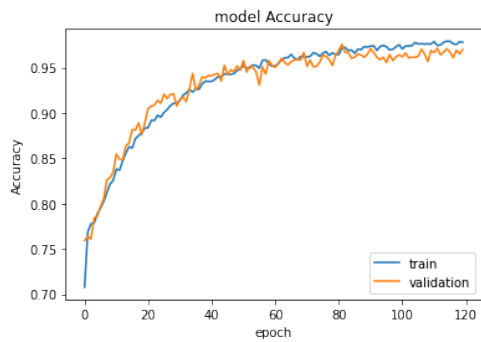


Fig. 9. Model accuracy graph

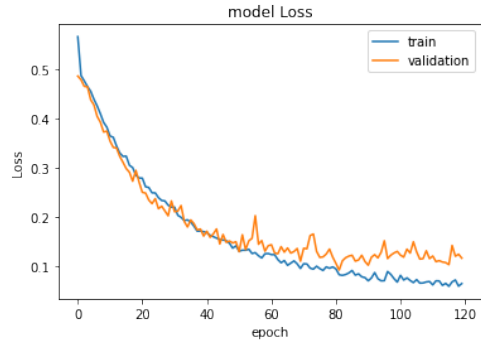


Fig. 10. Model loss graph

can be concluded from the figure that the architecture made can accurately guess as much as 1,850 data and only guess wrongly as much as 46 data.

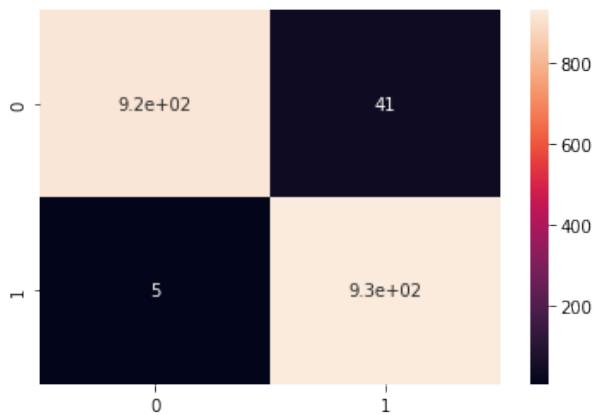


Fig. 11. Confusion matrix of proposed method

The architecture we use in this study is the Convolutional Neural Network (CNN) architecture to detect stroke by considering a collection of data regarding the factors that cause a stroke. We use CNN because this architecture has not been used by previous research using the same dataset. We take the data from the Kaggle website. The dataset consists of several features, namely gender, age, hypertension, heart_disease, ever_married, work_type, Residence_type, avg_glucose_level, BMI, smoking_status, and stroke, with the target feature being stroke. We chose the data in the form of stroke determinants or not. After analyzing the dataset, we noticed an imbalance in the dataset with the target attribute named stroke. Then we balance the data using the

Random Oversampling (ROS) technique. Simply this technique takes samples from the minority class, then duplicates them randomly and adds them to the training dataset. After the dataset is balanced, we try to make different CNN architectures to get the most optimal architecture results. Finally, we got the best CNN architecture in order of architecture which is Convolutional 1D (number of filters = 4, kernel size = 2, strides = 1, padding = same, activation = ReLU, pool size = 2, and strides = 1) twice ago using MaxPooling1D. After that, using Convolutional 1D (Conv1D), Flatten using the first 500 units dense layer and the second 10 units dense layer. We added a Dropout layer between the first and second dense layers to prevent overfitting the architecture. Finally, we use an output layer with dense 1 unit and sigmoid activation. For the optimizer, we use Adam, whose learning rate is 0.001, and the decay is 0.000001.

The dataset we use has several missing values in some features that cause the dataset to be imbalanced. We used the ROS technique to balance the number of datasets, in contrast to the study by Liu, T., et al. [11], which used a random forest architecture to relate missing values to the dataset before classification. In addition, the dataset that we use has 9,466 samples of data with ten crucial features and one target feature. It means that our dataset has more data than the dataset from the research of Singh, MS, & Choudhary, P. [7], which has as many as 5,888 data and a total of 600 features. Our research uses the Convolutional Neural Network architecture with a total of 12 layers compared to Yifeng Xie, et al. [8], which uses the Deep Neural Network architecture with a total of 12 layers.

TABLE I
SUMMARY OF PREVIOUS RESEARCHES

Article	Algorithm	Dataset	Accuracy
Liu, T., et al	Random Forest	Imbalanced Medical	71.67%
Yifeng Xie, et al	Deep Neural Network	ECG	85.82%
Singh, MS, & Choudhary, P.	Neural Network Classification	CHS	97.7%
Our proposed model	Convolutional Neural Network	Brain Stroke Dataset	98%

IV. CONCLUSION

This study was conducted to predict the possibility of a stroke based on the brain stroke dataset. The dataset consists of 4981 data with nine features in it. The dataset has a data imbalance, so the RandomOverSampler and StandardScaler functions are used to overcome it. The architecture used is Convolutional Neural Network with four convolutional layers and two hidden layers. Using the Adam optimizer, the learning rate is 0.001, the decay is 0.000001, and the model accuracy is 98 %. The model can be said to have a good performance already. This research contributes to the early prevention of possible stroke in the medical field. By using several variables regarding the factors that cause stroke, it can be seen that the architecture that has been made can predict the possibility of stroke in a person.

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